Supplementary S5 — Placement rules, amendments, operational decisions, and sensitivity of the grades

Generated from the rules file and the build output. The rules in Section 1 were frozen on 2026-10-24, before any placement was made. Section 2 gives the dated amendments, including those made after the grades were seen (A2, A3, A4). Section 3 gives the operational decisions. Section 4 gives the grade counts under the primary reading, under the rules as frozen, as first computed, and under each sensitivity scenario.

1. Rules as frozen

Frozen: 2026-10-24 (time and SHA-256 recorded in the project log). Any later change must be a dated amendment at the end of this file.

The phases are indicative age bands: Phase 1 ≈ 11–13 (ISCED 2), Phase 2 ≈ 14–16 (ISCED 2/3 boundary), Phase 3 ≈ 16–18 (ISCED 3). They form a **hypothesised learning progression** (Corcoran, Mosher & Rogat 2009), not a validated one.

Rules (applied to each competency × phase cell)

- **P1 Prerequisite.** A competency is not placed before the phase in which its mathematical and physics prerequisites appear in *standard* (non-advanced) secondary curricula. Evidence:
- Stadermann et al. 2019 (curricula of 15 countries; QP mostly in final-year electives, ages 17–19);
- Lehka 2021, Appendix Γ pp. 223–229 (Ukrainian programmes: probability and complex numbers only in grade-10 advanced mathematics; no matrices in any extract);
- curriculum facts reported in the included studies (screening.csv / evidence_matrix.csv).
- **P2 Representation demand.** Treatments are ordered as follows:
- verbal/visual (phenomenological) →
- probabilities and real-valued two-state descriptions tied to experiments (semi-quantitative) →
- complex amplitudes and 2×2 matrices (formal; an optional Phase-3 extension only).

Bouchée et al. (2022) give four reasons why quantum mechanics is difficult: (a) linking formalism to experience, (b) counter-intuitive phenomena, (c) the shift from deterministic to probabilistic thinking, (d) the limits of everyday language. They are used as **design constraints**, not as phases: - (b) and (d) → language and phenomena work first; - (c) → probabilistic reasoning is seeded early; - (a) → formalism only when anchored in an experiment (polarisation, Mach–Zehnder interferometer). - **P3 Age-matched evidence.** An included study that taught the topic to learners within the phase's age band and reported outcomes makes the cell *evidence-supported*. A study with learners in an adjacent band makes it at most *plausible*. Age or grade must be **reported** in the study. Ages inferred from grade are marked "inferred" and use the country's school system. - **P4 Consensus.** Expert or stakeholder consensus (Krijtenburg-Lewerissa et al. 2019; Merzel et al. 2024; Bitzenbauer et al. 2025; the printed stakeholder ratings in Lehka 2021, Tables 2.7–2.14) can support or oppose a placement but never places a topic on its own. - Bitzenbauer et al. 2025 act as a **ceiling**: advanced quantum-technology topics are treated conceptually, not mathematically, even in Phase 3. - Lehka's ratings are CFQT-derived items, so they are not independent of source S1; state this whenever they are used.

Confidence grades

- **H:** P1 met + ≥1 age-matched study with outcomes (P3) + no contrary consensus.
- **M:** P1 met + adjacent-age evidence or supporting consensus.
- **L:** only judgement or stakeholder support, or contrary evidence present.
- **Conj (conjecture):** no age-matched or adjacent-age evidence. The cell is a design proposal to be tested.

Evidence-type codes (matrix column `evidence_type`)

C = curriculum document; E = empirical classroom study with outcomes; A = adjacent-age empirical study; X = expert/stakeholder consensus; S = synthesis/review; J = authors' judgement.

Use of the Lehka (2021) thesis

Descriptive only, with page numbers: curriculum coverage, course feasibility, printed ratings. **Never as validation of the phases.** [One internal sentence, which names a separate manuscript, is redacted here for anonymous review.]

Grade definitions as frozen and as applied

The grade definitions frozen with the rules (above) were replaced by operational decision 1 (Amendment A1), written at 2026-10-25T20:53Z, before any cell was computed (the time-stamped placement log is published as S5, Part B). The frozen definitions overlapped (a cell with only supporting consensus was both M and L) and let consensus alone raise a cell to M, which rule P4 forbids; they also had no grade for cells in which the rules conflict.

Grade	As frozen (2026-10-24)	As applied (A1, 2026-10-25)
H	P1 met + at least one age-matched study with outcomes (P3) + no contrary consensus	Prerequisites met or not needed + at least one age-matched study that taught and tested the element and reported outcomes + no opposing consensus
M	P1 met + adjacent-age evidence or supporting consensus	As H but with an opposing consensus, or with adjacent-age evidence only
L	Only judgement or stakeholder support, or contrary evidence present	Contrary evidence (a null or negative outcome for an element that a study taught and tested)
Conj	No age-matched or adjacent-age evidence	No age-matched or adjacent-age evidence, whatever the consensus
U	(not defined)	Evidence exists, but the prerequisite and representation rules conflict, or no curricular source documents when a prerequisite is taught

No minimum sample size, design or reporting standard is set for H: one age-matched study that taught and tested the element and reported outcomes suffices. The matrix (S5_placement_matrix.csv) gives the records behind every grade, so each H can be weighed by its sample.

2. Amendments

Everything above this heading is the frozen text (full-file SHA-256 at freezing fd79e9529043392f…; SHA-256 of the part above this heading 1886be2930fcb463…). Amendments are appended below, dated, and say whether they were made before or after any placement result was seen.

A1 — operational decisions, fixed 2026-10-25T20:53Z, BEFORE any cell was computed (Section 3 below; recorded here 2026-10-26)

- **Grade precedence and an "undecided" grade.** The grade definitions above do not cover a placement whose rules conflict. A placed cell is graded, in this order: Conj if there is no age-matched or adjacent [taught+tested] evidence; **undecided** if P1 is not met but P2 admits the cell as an optional Phase-3 extension, or if no P1 source documents the curricular phase of a prerequisite; L if contrary evidence is present; H if P1 is met (or there is no prerequisite), age-matched evidence exists, and no consensus source opposes; M if the same holds but a consensus source opposes, or if the evidence is adjacent only.
- **No prerequisite** (a verbal or visual treatment that shows its own phenomenon) counts as P1 met. Sensitivity "na_neq_met" reports the alternative.

- **Overlap in the grade definitions.** "M … or supporting consensus" and "L: only judgement or stakeholder support" overlap with "Conj: no age-matched or adjacent-age evidence". Following P4 (consensus "never places a topic on its own") and the brief of the placement run of 2026-10-25, written before the matrix was filled, consensus or stakeholder support alone never raises a grade above Conj, and L is used only for contrary [taught+tested] evidence.
- Further operational decisions (representation matching, grade-only records, mixed samples, meanings of P4 support/neutral/oppose, adjacency) are items 2–12 of Section 3 below; the build script reports a sensitivity scenario for each.

A2 — correction of the P1 evidence statement, 2026-10-26, AFTER the matrix was built

- Lehka (2021) Appendix Γ is "Extracts from the curricula for grades 5–11" and spans pp. 223–233, not 223–229. Its mathematics extracts cover grades 10–11 only: the standard level shows coordinates and vectors; the advanced (pogliblenyi) grade-10 course shows probability theory (axioms, conditional probability) and complex numbers; the profile grade-11 course shows probability theory. Its only grade 5–9 extract is informatics (grade 8). Matrices appear in no extract.
- Because the appendix is a selection of extracts, it shows where these topics appear in the courses quoted. It does **not** show that probability is absent from standard courses, and it says nothing about lower-secondary mathematics. The frozen sentence "probability and complex numbers only in grade-10 advanced mathematics" is therefore corrected to: "in the extracts reproduced, probability theory appears in the advanced grade-10 and profile grade-11 courses and complex numbers in the advanced grade-10 course; no extract covers lower-secondary mathematics, and matrices appear in none".
- Effect on the matrix: none. The build already treated elementary probability in Phases 1–2 as "not established" (no P1 source documents it), so the cells that use it are undecided under A1.

Proposed 2026-10-26 after the matrix was built (all five adopted by the authors on 2026-10-26: see Amendment A3)

- AM-2: name a primary lower-secondary mathematics curriculum as an additional P1 source for elementary probability (relative frequency). This would be a post-results amendment; if adopted, the manuscript reports both grids.
- AM-3: allow an optional Phase-3 formal extension where an advanced or profile mathematics course supplies the prerequisite ("met, conditional").
- AM-4: state that the earliest admissible phase is a unit's entry.
- AM-5: state that Dirac notation with real coefficients is a real-valued two-state description (semi-quantitative).
- AM-6: decide whether this framework's own earlier placements satisfy P1 for units that depend on them (affects atom interferometry and atomic clocks).

A3 — five amendments adopted by the authors, 2026-10-26, AFTER the grades were seen

The authors adopted proposals AM-2 to AM-6 after seeing the grades. The grid reported in the manuscript applies A1–A3; the grid under A1–A2 only (the "as frozen" grid) is reported in Section 4 of this supplement alongside it. - **AM-2: a lower-secondary mathematics source for P1.** The standard national mathematics programme for grades 5–9 of the same system as the Lehka (2021) extracts is added as a P1 source (Ministry of Education and Science of Ukraine, *Matematyka. 5–9 klasy*, approved by order No. 804 of 7 June 2017). Its grade-9 algebra course, Topic 4, introduces "the frequency and probability of a random event". Elementary probability (relative frequency, probability of a random event) is therefore documented in a standard curriculum. Grade 9 is the last year of lower secondary (pupils aged about 14–15). By the school-level mapping applied to the other curricular grades (grade-8 informatics is Phase 1; grade 10, the first year of upper secondary, is Phase 2) it falls in Phase 1. By age it falls in Phase 2. The build counts the prerequisite as available from Phase 1, but no grade depends on the choice, because no Phase-1 placement has elementary probability as a prerequisite. For the same reason, naming the English (key stage 3) or US (Common Core grade 7) mathematics curriculum instead would give the same grades. Effect: 13 Phase-2 placements go from undecided to H and 4 to M. - **AM-3: optional formal extensions.** Where P2 admits an optional Phase-3 formal extension and an advanced or profile mathematics course supplies its prerequisite, the prerequisite counts as met, conditionally ("met, conditional"). Effect: complex amplitudes in Phase 3 go from undecided to Conj (no study taught them). - **AM-4: entry.** A unit's earliest admissible phase is its entry, and it is placed in every later phase. This states what the build already did; no grade changes. - **AM-5: Dirac notation.** Dirac notation with real coefficients counts as a real-valued two-state description (semi-quantitative). This states what the build already did; no grade changes. - **AM-6: the framework's own placements as prerequisites.** Where no curricular source documents a prerequisite by a phase, the prerequisite counts as met if this framework itself places, by that phase, a unit whose treatment teaches it, and that unit's own prerequisites are met there (no circularity). This applies only to energy levels (the energy-level diagrams of "Energy levels and line spectra", from Phase 2) and matter waves (the de Broglie wavelength, Phase 3 only). Effect: atomic clocks in Phase 2 go from undecided to Conj (no study taught them). - **Correction to A2.** A2 said the build already treated elementary probability in Phases 1–2 as "not established". In fact, the

prerequisite catalogue recorded it as "not met", citing the frozen sentence that A2 corrects. Both statuses gave "undecided" under A1, so no grade was affected. The catalogue now records "not established" with the corrected basis, and the "as frozen" grid uses that status. - **Full texts.** Four texts that had been fetched with a headless browser (R0016, R0026, R0048, R0103) were replaced by the authors' own browser downloads of the publisher PDFs. All four PDFs have the same DOIs, and every body paragraph and every extracted number matches. Locators into R0048 are now PDF page numbers (p.5 for the curricular statements on vectors, matrices, complex inner products and classical polarisation). No grade depends on how the texts were obtained. - **Totals.** With A3, of 123 placed cells, 46 are H, 35 M, 0 L, 42 Conj and 0 undecided. As frozen (A1–A2), they are 33 H, 31 M, 0 L, 40 Conj and 19 undecided. A3 changed 19 cells, all from undecided.

A4 — correction of the prerequisite catalogue, 2026-10-26, AFTER the grades were seen

- **What was wrong.** The P1 statuses that rest on Stadermann et al. (2019) were taken from the crosswalk extraction's summary ("QP courses mostly final year, ages 17–19") and were not checked against the source (the catalogue marked them "not re-read"). The citation check of the revised text (task T11) rendered Table V of the published article (p. 010130-10). In it, the compulsory physics course of grade 10, ages 15–16, in Bavaria covers Q5 wave–particle duality, Q6 matter waves, quantitative (de Broglie), and Q9 probabilistic or statistical predictions (marked ▲ by the source). Appendix B and p. 010130-13 state that physics is compulsory at that age for all Gymnasium (pre-university) students, 31 % of the 2017 cohort. No other column carries ▲.
- **Correction.** Under rule P1 as frozen (the earliest phase in which any standard, non-advanced curriculum among the sources documents a prerequisite), matter waves (P-DEB) and probability for measurement outcomes (M-PROB) are documented by Phase 2. The source states the ages, so no grade-to-age inference is needed. Later appearances (Ukraine grade 11; final-year courses in most systems) are recorded as dissent. All other Stadermann-based statuses were checked against Tables V and VI and are unchanged.
- **Cells re-specified.** Two cells had been specified as "not placed" in Phase 2 only because P-DEB was not met there: the particle in a box (K02-3) and atom interferometry, awareness (K08-4). They are now placed with the verbal-visual treatment of their Phase-3 cell. Treatments of cells that were already placed were not re-specified.
- **Effect, primary grid (2 cells):** K02-3-P2: — -> M; K08-4-P2: — -> Conj.
- **Effect under the rules as frozen (19 cells):** K01-1-P2: undecided -> H; K01-2-P2: undecided -> H; K01-3-P2: undecided -> H; K01-6-P2: undecided -> H; K01-7-P2: undecided -> H; K02-1-P2: undecided -> H; K02-2-P2: undecided -> H; K02-3-P2: — -> M; K02-5-P2: undecided -> M; K05-1-P2: undecided -> H; K05-2-P2: undecided -> H; K05-3-P2: undecided -> H; K06-2-P2: undecided -> H; K06-3-P2: undecided -> H; K07-1-P2: undecided -> M; K07-2-P2: undecided -> M; K07-4-P2: undecided -> H; K08-2-P2: undecided -> M; K08-4-P2: — -> Conj.
- **Consequence for A3.** With the correction, AM-2 changes no grade: the probability cells are already decided by the frozen sources. A3 now changes 2 cells relative to the corrected frozen grid: K02-4-P3: undecided -> Conj; K08-1-P2: undecided -> Conj.
- **Totals.** Primary (A1–A4): 125 placed: 46 H, 36 M, 0 L, 43 Conj, 0 undecided. Rules as frozen with the correction (A1, A2, A4): 125 placed: 46 H, 36 M, 0 L, 41 Conj, 2 undecided. As first computed (A1–A2, before the correction): 123 placed: 33 H, 31 M, 0 L, 40 Conj, 19 undecided, identical to the grid previously reported as frozen. Scenario noBY (primary without the correction): 123 placed: 46 H, 35 M, 0 L, 42 Conj, 0 undecided.
- The correction is reported in the manuscript and in Supplementary S5, together with the grid as first computed.
- **Source copy.** The PDF read for this correction (sha256 e0d788d0e9f7fb36…) is byte-identical to the authors' own local copy of the published article, which had been used for the crosswalk; the Table V marks did not survive that copy's text extraction (rotated table), which is how they were missed.

3. Operational decisions (fixed before any cell was graded)

The rules were applied literally. Where they were silent or ambiguous, the decisions below were fixed before any cell was computed; each has a sensitivity scenario in Section 4 where it could change a grade.

1. **Grade precedence and the undecided grade** (Amendment A1).
2. **Prerequisites are tools, not topics.** For each element and level of representation, the prerequisites are the mathematical tools and classical-physics concepts that the treatment uses but does not itself introduce. The earliest phase in which any standard (non-advanced, non-profile) curriculum among the sources documents a

prerequisite is that prerequisite's phase; later or non-standard appearances are recorded as dissent. Curricular grades are mapped to phases by school level (for example, a first year of upper secondary falls in Phase 2), and the mapping is marked as inferred.

3. **Conflicting rules.** Where the representation rule admits an optional Phase-3 formal extension but no standard curriculum supplies its prerequisite (complex amplitudes; matrices where no source documents them), the cell is placed and graded undecided, with the conflict stated. Amendment A3 (AM-3) later counted such a prerequisite as met, conditionally, where an advanced or profile course supplies it.
4. **Grades without ages.** A band is inferred from a reported grade only when the study names its country or school system; that system's grade-to-age mapping is used, and the band is marked as inferred and never reported as the study's age. Sensitivity scenarios S-A and S-B.
5. **Special samples.** A sample that mixes school and first-year university learners without separate school outcomes counts as adjacent evidence only (scenario O5alt counts it as age-matched). A study reporting several year groups credits each phase only with what the year group of that phase was taught and tested on.
6. **Meaning of consensus.** Support = the item met the source's own inclusion criterion. Neutral = below that criterion without a consensus to exclude. Oppose = an explicit rejection: high consensus with inclusion below half, or the expert study's ceiling against mathematical treatment of advanced quantum-technology topics. None of the consensus sources states that it concerns ages 11–13, so consensus support is recorded for Phases 2–3 only; an opposition applies to all phases. The stakeholder ratings in the thesis are derived from the workforce framework's items and are not independent of source S1. Scenario S-C reads every below-criterion item as opposed.
7. **Contrary evidence** must meet the same bar as support (taught and tested). Only a null or negative outcome, or a comparison in which the group without the element did better, counts. An element that a study taught but did not test is context, not contrary evidence.
8. **Elements without a topic code in the evidence matrix** are matched by stated rules (for example, real-valued two-state descriptions require a real-vector representation together with a two-state topic that was taught and tested, with real amplitudes). Programming has no topic code, so no study can support it; this is a stated limit.
9. **Placement.** The prerequisite rule sets the earliest phase, the representation rule sets the treatment in each phase, and the expert ceiling keeps quantum-technology topics conceptual. The rules paraphrase the expert study loosely. The study found support for teaching foundational topics conceptually rather than mathematically and judged more advanced quantum-technology topics unsuitable for secondary school; its panel stressed that the result should not limit future curricula. Keeping quantum-technology topics conceptual is therefore a design choice of this framework, applied as "no mathematical treatment of quantum-technology topics". An element is placed in every phase from its earliest admissible phase (Amendment A3, AM-4, states this in the rules).
10. **Awareness and formal entries** (the paper's Table 5) follow the two lowest proficiency levels of the workforce framework: awareness = what a technology does and why it matters; formal = an explanation of how it works, at any level of representation.
11. **Grid cells** carry one grade per listed element; grades are never merged upwards.
12. **Adjacency.** A study is age-matched for a phase if its (reported or inferred) age band intersects that phase's band, which overlap at 16; it is adjacent if it intersects only a neighbouring phase's band. A further check lists the Phase-1 placements whose adjacent evidence comes only from learners aged 15 or older.

4. Grade counts by scenario

45 content elements × 3 phases = 135 cells; 125 are placed. Evidence: 36 eligible studies with grade or age and outcomes, of which 25 taught and tested at least one element (22 samples after combining reports of the same sample).

Scenario	Placed	H	M	L	Conj	U	Phase 1 H/M/ Conj/U	Phase 2 H/M/ Conj/U	Phase 3 H/M/ Conj/U
Rules with Amendments A1 to A4 (the grades reported in the paper)	125	46	36	0	43	0	3/16/18/0	21/10/12/0	22/10/13/0
Rules as frozen (A1, A2) with the corrected prerequisite	125	46	36	0	41	2	3/16/18/0	21/10/11/1	22/10/12/1

Scenario	Placed	H	M	L	Conj	U	Phase 1 H/M/ Conj/U	Phase 2 H/M/ Conj/U	Phase 3 H/M/ Conj/U
catalogue (A4), without the post-results Amendment A3									
As first computed: A1 and A2, prerequisite catalogue before the A4 correction	123	33	31	0	40	19	3/16/18/0	8/5/10/18	22/10/12/1
A1 to A3 without the A4 correction (Bavarian grade-10 course not counted)	123	46	35	0	42	0	3/16/18/0	21/9/11/0	22/10/13/0
Without AM-2 (no lower-secondary mathematics source for elementary probability)	125	46	36	0	43	0	3/16/18/0	21/10/12/0	22/10/13/0
Without AM-3 (complex numbers not available from advanced or profile courses)	125	46	36	0	42	1	3/16/18/0	21/10/12/0	22/10/12/1
Without AM-6 (the framework's own placements do not count as prerequisites)	125	46	36	0	42	1	3/16/18/0	21/10/11/1	22/10/13/0
Age bands inferred from grades only where another included study or a curricular source states that system's grade-to-age mapping	125	40	35	0	50	0	1/17/19/0	20/8/15/0	19/10/16/0
No age bands inferred from grades	125	37	31	0	57	0	1/16/20/0	19/6/18/0	17/9/19/0
Topics below a consensus source's own criterion read as opposed rather than unsupported	125	5	77	0	43	0	0/19/18/0	3/28/12/0	2/30/13/0
	125	53	32	0	40	0	3/19/15/0	24/7/12/0	26/6/13/0

Scenario	Placed	H	M	L	Conj	U	Phase 1 H/M/ Conj/U	Phase 2 H/M/ Conj/U	Phase 3 H/M/ Conj/U
The mixed school and first-year university sample counted as age-matched rather than adjacent									
A treatment without prerequisites does not count as meeting the prerequisite rule	125	38	18	0	43	26	0/3/18/16	18/8/12/5	20/7/13/5
Studies that report no numerical outcome (R0064, a two-page extended abstract; R0094; R4098) are not counted as evidence. A reporting standard added on 2026-10-26, after the grades were known, as a sensitivity check only	125	42	34	0	49	0	1/16/20/0	19/10/14/0	22/8/15/0
H requires age-matched evidence from at least two independent samples (reports of the same sample counted once); a placement with one such sample is graded M. Added on 2026-10-26, after the grades were known, as a sensitivity check only	125	27	55	0	43	0	0/19/18/0	12/19/12/0	15/17/13/0
H requires at least one age-matched sample whose outcomes are reported per topic, item, lesson or key idea; a placement whose age-	125	37	45	0	43	0	1/18/18/0	16/15/12/0	20/12/13/0

Scenario	Placed	H	M	L	Conj	U	Phase 1 H/M/ Conj/U	Phase 2 H/M/ Conj/U	Phase 3 H/M/ Conj/U
matched samples report only aggregate scores is graded M. Outcome level classified by the language model from the S3 outcome summaries (S5_outcome_level. csv). Added on 2026-10-26, after the grades were known, as a sensitivity check only									
Age 16 counted only in Phase 3 (the indicative bands share it): samples whose youngest stated learners are 16 no longer count for Phase 2. Added on 2026-10-26, after the grades were known, as a sensitivity check only	125	45	35	0	45	0	3/16/18/0	20/9/14/0	22/10/13/0
Age 16 counted only in Phase 2: samples whose oldest stated learners are 16 no longer count for Phase 3. Added on 2026-10-26, after the grades were known, as a sensitivity check only	125	46	36	0	43	0	3/16/18/0	21/10/12/0	22/10/13/0

Samples behind the grades

Every sample that taught and tested at least one element and has an age band, with the full reference of each report, its design and size, the outcome as used in the matrix, whether that outcome is reported per topic (element) or only as an aggregate score, and the number of H placements of the primary grid that it supports as age-matched evidence. Reports of the same sample are combined. Outcome levels were classified by the language model from the S3 outcome summaries (S5_outcome_level.csv).

Sample	Report(s)	Design	n	Band	Outcome as used	Outcome level	H placements
R0222+R4109	R0222: Philipp Bitzenbauer (2021). Effect of an introductory quantum physics course using experiments with heralded photons on preuniversity students' conceptions about quantum physics. *Physical Review Physics Education Research* 17(2). doi:10.1103/physrevphyseducres.17.020103 R4109: Philipp Bitzenbauer, Jan-Peter Meyn (2020). A new teaching concept on quantum physics in secondary schools. *Physics Education* 55(5), 055031–055031. doi:10.1088/1361-6552/aba208	quasi-experimental pre/post with control group (traditional curriculum); acceptance survey (Teaching-experiment-type interviews); no control	280 surveyed (EG 150, CG 130); analysed 255 (EG 139 in 6 classes, CG 116 in 5 classes); 13 (7 male, 6 female)	P3 (inferred)	R0222: positive (EG > CG, d = 0.27/0.38; both gained) / R4109: positive (acceptance and paraphrasing positive)	element	4
R0227+R4134	R0227: Sara Satanassi, Elisa Ercolessi, Olivia Levrini (2022). Designing and implementing materials on quantum computing for secondary school students: The case of teleportation. *Physical Review Physics Education Research* 18(1). doi:10.1103/physrevphyseducres.18.010122 R4134: Sara Satanassi, Olivia Levrini (2025). Second quantum revolution: The progressive design of an approach to value its cultural and conceptual scope. *Physical Review Physics Education Research* 21(1). doi:10.1103/physrevphyseducres.21.010112	model of educational reconstruction; pilot post-only questionnaire, qualitative analysis, no control group; design-based research, three iterations; qualitative analysis; no control	approx. 20 per implementation (p3); 25 (2019) and 24 (2020) (p6); questionnaire respondents 14 of 22 (p15); almost 130 students over 5 rounds; teleportation questionnaire 14 of 22	P2;P3	R0227: positive, qualitative (14/14 respondents; "no statistical value") / R4134: positive, qualitative	element	5

Sample	Report(s)	Design	n	Band	Outcome as used	Outcome level	H placements
R0068+R0112	R0068: Avraham Merzel, Efraim Yehuda Weissman, Nadav Y. Katz et al. (2024). Mathematical structures in quantum physics education for high school students: Unveiling the power of Dirac notation for conceptual and problem-solving proficiency. *Physical Review Physics Education Research* 20(2). doi:10.1103/physrevphyseducres.20.020134 R0112: Efraim Yehuda Weissman, Avraham Merzel, Nadav Y. Katz et al. (2024). Keep it secret, keep it safe: teaching quantum key distribution in high school. *EPJ Quantum Technology* 11(1). doi:10.1140/epjqt/s40507-024-00276-4	retrospective qualitative content analysis of coursework/test responses; no pre-test, no control; retrospective content analysis of test/quiz responses and reflection questionnaire; no pre-test, no control	142 (item-level n varies, e.g. 125, 43); 7 interviewed; three 12th-grade classes; task-level n up to 32 (Q5), 13 (Q2-Q3); reflection questionnaire N=26 (two classes)	P3 (inferred)	R0068: positive (112/125 interpreted DN states) / R0112: positive (31/32, 24/32, 10/13)	element	10
R0571+R0016	R0571: Michele Darienzo, Angela M. Kelly, Dominik Schneble et al. (2024). Student attitudes toward quantum information science and technology in a high school outreach program. *Physical Review Physics Education Research* 20(2). doi:10.1103/physrevphyseducres.20.020126 R0016: Robert De La Cruz, Angela M. Kelly, Tzu-Chieh Wei et al. (2026). Academic and demographic predictors of high school student knowledge of classical physics, quantum physics, and quantum	mixed methods explanatory sequential; pre/post with non-equivalent control (pretest comparison); interviews; pre/post single group with paired t-tests, ANOVA/ANCOVA by site, year, gender, ethnicity, coursework, grade; no control	77 treatment (60 paired pre/post in t-test, d.o.f. 59); 65 control; 12 interviewed; 209 (paired analyses n=200)	P2;P3	R0571: not coded / R0016: positive (inventory 36.7 -> 62.9 %, d = 1.65)	aggregate	19

Sample	Report(s)	Design	n	Band	Outcome as used	Outcome level	H placements
	computing in university-based outreach. *EPJ Quantum Technology* 13(1). doi:10.1140/epjqt/s40507-026-00515-w						
R4107	R4107: U. Scotti di Uccio, Arturo Colantonio, Silvia Galano et al. (2020). Development of a construct map to describe students' reasoning about introductory quantum mechanics. *Physical Review Physics Education Research* 16(1). doi:10.1103/physrevphyseducres.16.010144	two-group comparison (TLS vs textbook), post-test; construct-map validation with Rasch	408 (TLS 200, textbook 208)	P3	R4107: positive (TLS 0.27 vs -1.13 logit)	aggregate	4
R0026	R0026: Carsten Albert, Moritz Förster, Gesche Pospiech (2025). Developing a quantum physics curriculum for lower secondary education: insights into the design of a Spin First teaching concept and first empirical findings. *EPJ Quantum Technology* 12(1). doi:10.1140/epjqt/s40507-025-00436-0	design-based research; formative evaluation by one-to-one Teaching Experiments; no control; self-selected, positively biased sample	11 (6 female, 5 male)	P2	R0026: positive (all key ideas below the 0.5 success threshold)	element	10
R0029	R0029: Vijit V. Nautiyal, Erika Eunice P. Salvador, Xavier Romy O. Braña et al. (2025). Enhancing high school students' understanding and attitude towards quantum mechanics through discipline-culture framework and cognitive apprenticeship. *EPJ Quantum Technology* 12(1). doi:10.1140/epjqt/s40507-025-00407-5	single-group pre/post with qualitative journals; no control	20 (11 male, 9 female)	P2;P3	R0029: positive (QPCS 12.75 -> 16.05/25, p = .03)	aggregate	14

Sample	Report(s)	Design	n	Band	Outcome as used	Outcome level	H placements
R0040	R0040: Marjona Toshboyeva (2026). QUANTUM PHYSICS ORIGINS TEACHING METHODOLOGY FOR SECONDARY SCHOOLS: A PEDAGOGICAL AND STATISTICAL ANALYSIS. *Journal of Multidisciplinary Sciences and Innovations*.	quasi-experimental pre/post with control class	52 (26 control, 26 experimental)	P3 (inferred)	R0040: positive (EG > CG; t = 6.82)	aggregate	4
R0048	R0048: Giacomo Zuccarini, Claudio Sutrini, Maria Bondani et al. (2024). Teaching quantum information science to secondary school students with photon polarization and which-path encoding. *EPJ Quantum Technology* 11(1). doi: 10.1140/epjqt/s40507-024-00287-1	model of educational reconstruction; preliminary evaluation: worksheet analysis + pre-post attitude item (unpaired t-test); no control	27 (worksheet sub-items analysed for 11-12 students; post survey 25/27)	P3 (inferred)	R0048: positive (gate design 20/27)	element	6
R0056	R0056: Megan Ivory, J. T. Balewski, Alisa Bettale et al. (2025). QCaMP: A 4-Week Summer Camp Introducing High School Students to Quantum Information Science and Technology. *IEEE International Conference on Quantum Computing and Engineering (QCE)*, 29–38. doi:10.1109/qce65121.2025.20510	single-group pre/post (knowledge items + attitudes, t-tests); no control	41 enrolled (24 + 14 + 3); 40 assessment responders	P2;P3 mixed	R0056: positive (all lessons gained; most > 80 % post)	element	0
R0064	R0064: Zeynep Akdemir, Muhsin Menekşe (2024). Designing Quantum-Infused NGSS-Aligned Science Curriculum to Support Learning and Engagement of Middle School Students. *Proceedings.*, 2319–2320. doi:10.22318/icls2024.358019	design-based research, final phase: single-group pre/post (paired t-tests); no control	873 (students with both pre and post measures)	P1;P2 (inferred)	R0064: positive (significant gain, moderate-large effect; no numbers)	aggregate	5

Sample	Report(s)	Design	n	Band	Outcome as used	Outcome level	H placements
R0094	R0094: Vijit V. Nautiyal, Nadya Rizk, Frances Quinn (2026). Reconstructing the history of light and the birth of photon: Integrating nature of science into introductory quantum mechanics. *Eurasia Journal of Mathematics Science and Technology Education* 22(8), em2890–em2890. doi: 10.29333/ejmste/19051	design/narrative paper with pilot; outcome only summarised		P2 (inferred)	R0094: weak-positive (positive NOS shift, no n or numbers)	element	1
R0238	R0238: Kristóf Tóth, Marisa Michelini, Philipp Bitzenbauer (2024). From light polarization to quantum physics: Supporting lower secondary school students' transition from gestalt to functional thinking. *Eurasia Journal of Mathematics Science and Technology Education* 20(6), em2449–em2449. doi: 10.29333/ejmste/14587	evaluation study, pre/post within the sequence (pre after lesson 3), single group, no control	51 (25 female, 26 male)	P2;P3	R0238: positive (QM-thinking answers rose on all tasks)	element	12
R0251	R0251: Fabian Hennig, Kristóf Tóth, Joaquin Marc Veith et al. (2024). Mathematical sense making of quantum phenomena using Dirac notation: its effect on secondary school students' functional thinking about photons. *EPJ Quantum Technology* 11(1). doi: 10.1140/epjqt/s40507-024-00274-6	two cluster-randomised field studies: pre/post with control group (study 1); post-only cognitive-load comparison (study 2)	study 1: 133 (IG 67 in 3 classes, CG 66 in 4 classes); study 2: 136 (IG 71, CG 65; 7 classes)	P2;P3	R0251: positive (both groups improved, $p < .01$)	aggregate	10
R0483	R0483: Marisa Michelini, Alberto Stefanel, Kristóf Tóth (2022). Implementing Dirac Approach to Quantum Mechanics in a Hungarian Secondary	pilot implementation with pre/post test in two groups; no control	24 (A: 5, B: 19); interviews 5	P3	R0483: mixed (significant gain only for group A, n = 5)	element	6

Sample	Report(s)	Design	n	Band	Outcome as used	Outcome level	H placements
	School. *Education Sciences* 12(9), 606–606. doi:10.3390/educsci12090606						
R0493	R0493: Padmanabh Kaushik, Nam P. Vu, Crystal Yeung et al. (2026). Quantum Chess as a Pedagogical Tool for Teaching Quantum Information Science in High Schools. *Journal of Chemical Education* 103(7), 3971–3980. doi:10.1021/acs.jchemed.5c00836	single-group pre/post survey; no control; very small sample	19 participants; 16 pre-survey; 13 post (10 complete pre+post)	P2;P3 (inferred)	R0493: positive (8/10 superposition definitions improved)	element	8
R0933	R0933: Philipp Bitzenbauer, Sergej Faletič, Marisa Michelini et al. (2024). Design and evaluation of a questionnaire to assess learners' understanding of quantum measurement in different two-state contexts: The context matters. *Physical Review Physics Education Research* 20(2). doi:10.1103/physrevphyseducres.20.020136	instrument development: qualitative pilot, expert survey, quantitative evaluation with group comparison (post-only)	study I: 120 (97 high-school + 23 prospective teachers); study III: 201 (BS 11, POL 84, DW 42, rest traditional)	P2;P3	R0933: positive (two-state approaches above traditional; post-only)	aggregate	2
R4098	R4098: Massimiliano Malgieri, Pasquale Onorato, Anna De Ambrosis (2017). Test on the effectiveness of the sum over paths approach in favoring the construction of an integrated knowledge of quantum physics in high school. *Physical Review Physics Education Research* 13(1). doi:10.1103/physrevphyseducres.13.010101	case study of a teaching-learning sequence; post-instruction questionnaire; comparison with literature, no control	18 in class; 14 took the final research test	P3 (inferred)	R4098: mixed (duality "may be effective"; uncertainty "less clear cut")	element	4
R4101	R4101: Francisco Savall Alemany, Jenaro	quasi-experimental: TLS cohorts vs conventional-		P3		element	3

Sample	Report(s)	Design	n	Band	Outcome as used	Outcome level	H placements
	Guisasola, Sergio Rosa Cintas et al. (2019). Problem-based structure for a teaching-learning sequence to overcome students' difficulties when learning about atomic spectra. *Physical Review Physics Education Research* 15(2). doi:10.1103/physrevphyseducres.15.020138	teaching cohorts, pre/post questionnaire	74 experimental (5 cohorts, 4 years) + 67 control (4 cohorts)		R4101: positive (TLS 59-66 % vs control 7.5-9 %)		
R4121	R4121: Kim Krijtenburg-Lewerissa, Hendrik Jan Pol, Alexander Brinkman et al. (2021). Prior knowledge of potential energy and the understanding of quantum mechanics. *Physics Education* 57(2), 025012–025012. doi:10.1088/1361-6552/ac3d3a	quasi-experimental: classes with energy module vs control classes; pre/post tests	391 (experimental 234 in 13 classes; control 157 in 11 classes)	P3 (inferred)	R4121: positive (EG > CG, small effects)	element	0
R4133	R4133: Luiza Vilarta Rodriguez, Jan T. van der Veen, Ton de Jong (2025). Role of analogies with classical physics in introductory quantum physics teaching. *Physical Review Physics Education Research* 21(1). doi:10.1103/physrevphyseducres.21.010108	quasi-experimental two-condition comparison (class-level assignment), post-test only, covariate adjustment	112 in classes; 89 took post-test (43 analogy, 46 no analogy); 56 matched to central exam	P2;P3	R4133: comparison only (no-analogy > analogy; no gain measure)	element	4
R0125-Y8	R0125-Y8: Tejinder Kaur, Magdalena Kersting, D. G. Blair et al. (2024). Developing and implementing an Einsteinian science curriculum from years 3–10: A. Concepts, rationale and learning outcomes. *Physics Education* 59(6), 065008–065008. doi:	mixed-methods MER trials; pre/post per year group; no control	315 across Years 3-10; Year 9 Quantum world 42; Year 8 Einsteinian energy 22	P1	R0125-Y8: positive (77 % E=hf item)	element	1

Sample	Report(s)	Design	n	Band	Outcome as used	Outcome level	H placements
	10.1088/1361-6552/ad66a7						
R0125-Y9	R0125-Y9: Tejinder Kaur, Magdalena Kersting, D. G. Blair et al. (2024). Developing and implementing an Einsteinian science curriculum from years 3–10: A. Concepts, rationale and learning outcomes. *Physics Education* 59(6), 065008–065008. doi: 10.1088/1361-6552/ad66a7	mixed-methods MER trials; pre/post per year group; no control	315 across Years 3-10; Year 9 Quantum world 42; Year 8 Einsteinian energy 22	P2	R0125-Y9: positive (50 % EP-consistent photons; 76 % photons-waves)	element	3

5. Files

- `placement_matrix.csv`: every element × phase cell, with treatment, prerequisite status and basis (sources with page or line numbers), the studies used (with their bands and outcome direction), consensus basis, grade, and the grade under each scenario.
- `placement_entry.csv`: the earliest awareness and formal entry of each element.
- `placement_sensitivity.csv`: the grade of every cell under each scenario of Section 4.
- `placement_units.csv`: the content elements of the paper's Table 3, with their domains.
- `placement_samples.csv`: the samples behind the grades (the table in Section 4).
- `outcome_level.csv`: for each evidence record, whether its outcomes are reported per topic or only as aggregate scores, with the basis quoted from S3 (classified by the language model).
- Three grids: as printed (the paper's Table 4, Amendments A1–A4); under the rules as frozen with the corrected catalogue (A1, A2, A4); and as first computed (A1–A2, before the A4 correction).

Supplementary S5, Part B — Time-stamped log of the placement run (operational decisions O1–O13 and build steps)

Published copy of the project file `placement.md`; SHA-256 of the original: `b393c5cc818971d02b642242f160cb4c699c070658b8f03040e50d0cfa5d7e9c`. 18 redactions, marked ⟦…⟧, replace internal file names, internal role names, or references to a separate manuscript, and local times are given in UTC; nothing else is changed. Times are UTC. The log shows that the operational decisions (O1–O13) were written at 20:53Z and corrected at 21:05Z, before the first build at 21:11Z.

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placement agent (T6) — log

Agent: `placement` (Claude, model ID claude-opus-5-5). Brief: ⟦task brief⟧. Rules: `evidence/placement_rules.md` (FROZEN). Restart protocol: re-read this log; skip unit_ids marked DONE; resume at the "next" of the last entry.

Log

- 2026-10-25T20:32:58Z | unit=U00-start | result: rules sha256 = fd79e9529043392fbdefc16855d7d914ed52a9195a3032fc65b923998f12b1e8 (prefix fd79e9529043392f OK). Read brief, rules, crosswalk_clusters.csv, crosswalk_decisions.md (incl. Addenda B, C), paper_q.tex tab:competency_domains / tab:framework_pathway_map / tab:twodim (read-only), evidence_matrix.csv eligible 36 rows. | files: ⟦this log⟧ | next: U01 read P1/P4 sources (evidence/sources, crosswalk_extraction S5 Tables III-V, Lehka App. Γ + Tables 2.7-2.14)
- 2026-10-25T20:41:01Z | unit=U01-sources | DONE. Read: S5 bitzenbauer2025.txt (Table II levels p.020157-6; Tables III-V p.020157-10..12 give Cns + Include% ONLY, no per-topic level -> brief's statement that III-V give levels is inaccurate; only the 10 Table II topics have level %); crosswalk_extraction S2/S3/S5/S6 rows (S6 = Stadermann Table VI counts via T5; Stadermann full text NOT cached in this project — T5 read it from ⟦a folder outside this project⟧, which I must not open -> P1 uses T5's extraction + rules summary); Lehka thesis App. Γ = PDF pp.223-233 (printed = PDF page numbers; not 223-229 as rules say), Tables 2.7-2.14 located (lines 4096-5093); Lehka gives no grade-to-age mapping; role-F abstracts (R0004, R0011, R0042, R0024, R0353); ⟦the main model session's notes (record of Semerikov et al. (2026),⟧ Table 3: superposition P2, entanglement P3, matrices P2; levels phenomenological/conceptual/analytical). | files: none written | next: U02 advisor check of operationalisation, then evidence/placement_units.csv
- 2026-10-25T20:53:22Z | unit=U02-operationalisation | DONE (advisor check done before writing). Decisions fixed BEFORE any cell is computed; each is applied literally and its sensitivity is reported by build.py. | files: ⟦this log⟧ | next: U03 evidence/placement_units.csv

Operationalisation and proposed amendments (dated 2026-10-25; placement_rules.md NOT edited)

Rules read literally. Where they are ambiguous, the most literal reading is applied and an amendment is proposed to ⟦the main model session⟧ (AM-n). None is applied silently.

O1 Grade precedence (evaluated in this order for every placed cell). 1. Unit not placed in the phase → grade "—". 2. No age-matched and no adjacent [taught+tested] support → **Conj** (brief: "whatever the consensus says"; this overrides the rules' "M: … or supporting consensus"). 3. P1 = "not met" and the cell is placed only because P2 allows an optional Phase-3 extension (P1-vs-P2 conflict) → **undecided**. 4. P1 = "not established" (no P1 source documents the phase of a prerequisite) → **undecided**. 5. Contrary evidence present (a [taught+tested] record at matched or adjacent band reporting no gain/decline, or a comparison where the group without the unit/representation did better) → **L**. 6. P1 met or n.a. + ≥1 age-matched record + no P4 oppose → **H**. 7. P1 met or n.a. + ≥1 age-matched record + P4 oppose → **M** (age-matched counts a fortiori as "adjacent-age evidence"). 8. P1 met or n.a. + adjacent-only evidence → **M**. - AM-1 (proposed): state in the rules that "n.a." (no prerequisite) satisfies "P1 met" in the grade definitions; that undecided is a grade; and the precedence above. Literal reading applied meanwhile = n.a. treated as vacuous compliance (a prohibition with nothing to prohibit is met). Sensitivity: n.a. ≠ met. - P4 support never changes a grade under item 2 (evidence is required anyway); it is recorded only.

O2 P1 = prerequisites, not topic presence. Each unit × representation level lists the mathematical tools and the classical-physics concepts the treatment USES as an explanatory model but does not introduce (prereq catalogue: tools/placement/prereq_catalogue.csv). Verbal-visual treatments that show their own phenomenon have none → P1 "n.a.". - EDSP = earliest documented standard-curriculum phase of a prerequisite. Combining countries: earliest phase documented in ANY standard (non-advanced, non-profile) curriculum by a P1 source; later or non-standard appearances are recorded as dissent. - A treatment whose own content is listed in a standard curriculum at phase X has its prerequisites available at X (e.g. Stadermann Q9 "probabilistic predictions" in QP courses → probabilistic description of measurement available by Phase 3). - P1 status per cell: met (all prereqs EDSP ≤ phase) / not met (some EDSP > phase, or documented only in non-standard tracks) / not established (some prereq with no EDSP and none later) / n.a. (no prereqs). - Grade → phase mapping for curricula: by school level (Ukrainian grade 10 = first year of upper secondary = ISCED 2/3 boundary → Phase 2; grade 11 → Phase 3). Lehka states no ages; every such mapping is marked "inferred (school level)". - Facts already read: real vectors standard grade 10 (Lehka p.223); probability THEORY only advanced grade 10 (p.223) and profile grade 11 (pp.224–225); complex numbers only advanced grade 10 (pp.223–224); matrices absent (pp.223–233); classical interference, diffraction, photon, photoeffect, solar cells, duality of light, laser coherence in standard grade 10 physics (p.226); Bohr levels, emission/absorption, line spectra, laser principle, de Broglie in standard grade 11 (p.227); polarisation, Michelson interferometer only profile grade 11 (p.228); uncertainty, potential well, tunnelling, spontaneous/stimulated emission, Pauli only profile grade 11 (p.229); SI units, measurement uncertainty profile grade 10 (pp.227–228); binary coding and programming informatics grade 8 (pp.229–230). App. Γ spans pp.223–233 (rules say 223–229). No Lehka extract covers grades 5–9 physics or maths → no P1 source covers lower-secondary prerequisites. - Included-study curriculum facts: R0048 l.172/l.176 (Italy, science-oriented upper secondary: classical polarisation scheduled; vectors extensively; matrices peripheral; no complex inner product) [headless]; R0251 p6 (Germany: matrix multiplication not in school maths); R4107 p3 (Italy: chemistry grades 9–12 introduce Bohr model, orbitals, spin; QM in 13th grade); R4105 p4, R4103 p4 (NL last year VWO 17–18: duality, photoeffect, uncertainty, 1D well, H atom, tunnelling); R0103 §4.1 [headless], R0144 p3, R0172 p7 (Germany: mandatory introductory QP course before/at grades 11–12); R0222 (Bavaria grade-12 traditional curriculum control group taught the probability interpretation). R4134 p6 describes its volunteer sample ("typically had ... matrices, vectors, complex numbers") — a sample description, not a curriculum statement → recorded as dissent only. - Elementary probability (relative frequency, simple fractions) has NO P1 source for Phases 1–2 → "not established" there. AM-2 (proposed): authors supply a lower-secondary maths curriculum source, or the rules state that elementary probability counts as available from Phase 1. Literal reading applied meanwhile.

O3 P1 vs P2 for complex amplitudes and matrices. P2 allows them as an optional Phase-3 extension; P1 (standard curricula: Lehka advanced-only/absent; R0048; R0251) is not met in any phase. Cells are placed (P2) but graded **undecided** with the conflict stated. AM-3 (proposed): "An optional Phase-3 formal extension is permitted where a non-standard (advanced/profile) mathematics course supplies the prerequisite; P1 is then recorded as 'met (conditional)'." Sensitivity: grade under AM-3.

O4 Grade-only records. Primary (rules P3 + brief): infer a band only when the paper names the country/school system; use that country's system; mark the band "inferred" and never write it as the study's age; cite a corroborating included record where one maps that system. No country named → no inference (the record can still act through an overlap group whose other member names it). Sensitivity S-A: infer only when another included record or a P1 source states the grade-to-age mapping for that system (advisor's middle reading). Sensitivity S-B: no inference at all.

O5 Special samples. Mixed samples with school outcomes not separated (R0056: 5/40 college freshmen; eligibility ≥16) count as ADJACENT only (at most plausible) in the primary reading; sensitivity: age-matched. R0933 study III (school outcomes separate, 16–18) counts as age-matched. R0125 is split by year: Year 8 (age 13 by the paper's mapping, p12 l.13–37) credits Phase 1 only with what Year 8 tested (E=hf photon energy); Year 9 (age 14, p12 l.38–58) credits Phase 2 with photons, photons-and-waves (duality), solar panels.

O6 P4 meanings. Support = the item met that source's own criterion (S3 selected for round 3; S4 endorsed by its cut-off; S5 Table II both criteria; Lehka A ≥ 120.8, p.88). Neutral = below criterion without consensus (S5 Tables IV–V: "no consensus was reached"; S3/S4 not selected; Lehka not selected). Oppose = explicit rejection: S5 Table III items with Cns ≥ 0.75 AND Include < 50 % (S5's own reading that high consensus with low inclusion is agreement to exclude, p.020157-7), and the S5 ceiling against any mathematical-level treatment of a Table V quantum-technology topic. S5 levels are used only for its 10 Table II topics (Tables III–V print Cns and Include % only). Scope: S3 "secondary schools", S4 "high school", S5 "secondary school" benchmarked against upper-secondary curricula (§VII.A), Lehka lyceum grades 10–11 → none states lower secondary/ages 11–13, so P4 support is recorded for Phases 2–3 only; an opposition ("not needed at secondary school") applies to all phases. Lehka ratings are CFQT-derived items, not independent of S1 — stated in every cell that uses them. Sensitivity S-C: below-criterion counted as oppose (cells moving H→M reported). - Lehka printed values checked against Tables 2.7–2.14 (pp.89–100): ⟦the main model session's notes⟧ is right on 7.04 = 92, 7.01 = 122, 7.08 = 125, 2.05 = 113, sensing share 10.33 % (p.101), but omits that 2.01 "knowledge of the basics of complex-number theory" = 144 and 2.02 "linear algebra" = 156 and 2.10 "unitary matrices" = 132 were SELECTED (p.91).

O7 Contrary evidence must clear the same [taught+tested] bar as support. Outcome direction is coded per record (and per topic where the record separates it) from outcome_result: positive / weak-positive (gain but low post level) / mixed / null-negative. Only null-negative or "comparison group without the unit did better" is contrary. R0251's Dirac result (Dirac only [taught]) is context, not contrary evidence.

O8 Units without a topic code match by rule: real-valued two-state vectors = representation contains "real-vector" AND a two-state topic [taught+tested] (polarisation, single photons, interference/MZI, superposition, qubit, Dirac notation, other:double-well, other:state circle) AND notes say "amplitudes: real" or the representation lacks "complex"; qubit measurement = qubit AND measurement and probability both [taught+tested]; entanglement as resource = entanglement [t+t] AND (qubit or gates/circuits or other:quantum teleportation) [t+t]; QC capabilities = other:basic idea of quantum computers; noise/errors = other:decoherence or other:noise and error correction; programming has no code → no record can support it (stated as a limit).

O9 Placement generation. P1 sets the floor, P2 sets the representation of each phase (Ph1 verbal-visual + qualitative probabilistic reasoning; Ph2 semi-quantitative; Ph3 semi-quantitative + optional formal), P4 (S5 ceiling) caps QT topics at conceptual. A unit is placed in every phase from its earliest admissible phase; the treatment text is derived from the P2 level, not from the R1 grid. Units formal by nature (complex amplitudes, Dirac notation, 2×2 matrices) are "—" where P2/P1 exclude them. AM-4 (proposed): the rules constrain but do not say a unit should be placed as early as admissible; state that the earliest admissible phase is the entry and that it is a conjecture unless graded otherwise.

O10 entry_type follows tab:twodim, not P2: awareness = CFQT A1 "what it does and why it matters"; formal = A2 "explains how it works" (at any representation, including conceptual circuits). Earliest awareness phase = first placed phase; earliest formal phase = first phase whose treatment explains how it works.

O11 Grid cells: one grade letter per listed unit; never merged upwards; if a single letter is forced, the lowest.

O12 Adjacency. Age-matched = the record's (stated or inferred) band intersects the phase band (bands overlap at 16). Adjacent = the band intersects a neighbouring phase band only. A record spanning P2;P3 is age-matched for both and adjacent to P1. Sensitivity S-D: Phase-1 cells whose adjacency rests only on records whose youngest stated learners are ≥15.

O13 Sources not re-read. Stadermann 2019 and Krijtenburg-Lewerissa 2019 full texts are not in this project (T5 read them from ⟦a folder outside this project⟧, which I may not open); their facts are taken from crosswalk_extraction.csv (S3, S6 rows) and the rules' summary ("QP mostly in final-year electives, ages 17–19; only Bavaria compulsory at 15–16"), labelled "not re-read". Semerikov et al. (2026) placements are ⟦"the main model session's notes⟧, not checked against Table 3". Request to ⟦the main model session⟧: confirm both. - 2026-10-25T21:01:38Z | unit=U02b-O2-clarification | Before any cell: (1) Probability follows the rules' own P1 text ("probability and complex numbers only in grade-10 advanced mathematics", Lehka pp.223–225), so M-PROB is "not met" in standard curricula for Phases 1–2 and "met" in Phase 3 (Stadermann Q9 [not re-read]; R0222 control). Semi-quantitative Phase-2 cells that use probabilities are therefore a P1–P2 conflict → undecided (O1 step 3). AM-2 (elementary probability ≠ probability theory; or advanced-track availability counts) is the proposed resolution; its grades are reported as sensitivity. (2) P1 physics prerequisites = classical or quantum concepts documented in P1 curriculum sources that a treatment USES as its explanatory basis. A unit's own content is not its own prerequisite. Quantum concepts that no curriculum source lists (superposition, qubit) are framework content: dependent units follow the framework sequence (J), not P1. AM-6 (proposed): should content this framework places earlier satisfy P1 for dependent units (e.g. matter waves → atom interferometry)? Literal reading meanwhile = no; P1 uses standard curricula only. (3) The S5 ceiling (O6/O9) caps quantum-technology units (S5 Table V topics: gates, BB84, QRNG, cryptography comparison, sensors, atom interferometers, SI units, imaging, photon sources/detectors, fibres, vacuum, flash memory, microelectronics) at conceptual use: no calculations; formal-optional cells are not placed for them. (4) Dirac notation: P2 does not name it; with real coefficients it is read as a real-valued two-state description (semi-quantitative), and with complex coefficients as formal. AM-5 (proposed): state this in P2. | files: ⟦this log⟧ | next: U03 evidence/placement_units.csv - 2026-10-25T21:05:39Z | unit=U03-units | DONE. evidence/placement_units.csv written by tools/placement/00_units.py: 45 units (K01 7, K02 6, K03 4, K05 3, K09 1, K06 3, K07 4, K08 4, K10 3, K12 4, K13 3, SOC 3). sha256 466486a1fa727a4ec7d6599eada658ad8e3e429e0c6a7c6da02b53c68bef5007. Units added for reviewer cases: K08-4 atom interferometry (case a), K07-1/K07-2 BB84 concept vs protocol split (case b), K02-4/K02-5/K02-6 (cases b, d), K01-1/K01-6 (case c). **O1 order corrected before the first build** (append-only correction): 1 not placed → "—"; 2 P1–P2 conflict (P2 assigns this representation to this phase, P1 not met) → undecided; 3 P1 not established → undecided; 4 no age-matched/adjacent evidence → Conj; 5 contrary evidence → L; 6 H; 7 M (age-matched + P4 oppose); 8 M (adjacent only). Reason: when P1 compliance is in conflict or unknown, whether the cell is placed at all is undecided, so "Conj" (a design proposal) would presume a placement the rules have not made. P1 "not met" with no P2 mandate (e.g. matter waves for atom interferometry in Phases 1–2) → not placed. | files: tools/placement/00_units.py, evidence/placement_units.csv | next: U04 prereq_catalogue.csv, lehka_ratings.csv, cell_spec - 2026-10-25T21:07:37Z | unit=U04-spec | DONE. tools/placement/01_catalogues.py → prereq_catalogue.csv (14

prereqs; sha256 709fc8af…) and lehka_ratings.csv (64 printed items, Tables 2.7–2.14 pp.89–100; sha256 b73ba28c…). tools/placement/02_cells.py → cell_spec.csv (135 cells; sha256 c8e6eae9…). Not-placed cells: K02-1/K02-2/K02-5/K07-2 Phase 1 (P2), K02-4/K02-6 Phases 1–2 (P2), K02-3 and K08-4 Phases 1–2 (P1: matter waves). Spec written before any evidence lookup per cell (the evidence matrix was read in U00 as orientation only). | files: tools/placement/{01_catalogues.py, 02_cells.py,prereq_catalogue.csv,lehka_ratings.csv,cell_spec.csv} | next: U05 build.py - 2026-10-25T21:11:36Z | unit=U05-build | First build run. Two bugs found and fixed before using any result: (i) eligibility let in R1487 (ft_flag: no learner outcome data) → now identical to T2b summarize.py has_out() + i3 met (36 ids, equal to the JSON list); (ii) contrary detection matched "no gain measure" in R4133's comparison label → now only an explicit null/negative/contrary category counts (O7). Primary result (123 placed cells of 135): H 33, M 31, L 0, Conj 40, undecided 19 (P1: H3 M16 Conj18; P2: H8 M5 Conj10 U18; P3: H22 M10 Conj12 U1). Sensitivities in tools/placement/build_report.txt and evidence/placement_sensitivity.csv. | files: tools/placement/build.py, evidence/placement_matrix.csv, evidence/placement_entry.csv, evidence/placement_sensitivity.csv, drafts/placement_grid.tex, drafts/placement_grid_AM2.tex, tools/placement/build_report.txt | next: U06 audit key cells (reviewer cases, Phase-1 H cells, headless), then test-compile the grid - 2026-10-25T21:13:57Z | unit=U06-audit+compile | DONE. Audited the reviewer-case cells and the Phase-1 H cells by hand against evidence_matrix rows. Fixed labels (S5 Table II cited for both-criteria items; headless flag on a P1 basis only where the status depends on R0048). Report now lists cells whose P3 status changes without the 4 headless texts. drafts/placement_grid.tex test-compiled in a scratch cas-sc document (scratchpad/gridtest/test.tex, with natbib and colortbl as in the manuscript): 0 errors, 0 overfull boxes, the grid fits one page at scriptsize. | files: tools/placement/build.py, evidence/placement_matrix.csv, tools/placement/build_report.txt | next: U07 drafts/twodim_check.md, placement_cases.md, placement_section.md - 2026-10-25T21:15:37Z | unit=U07a-twodim | DONE. drafts/twodim_check.md written. tab:twodim matches on rows 1 (PQC), 6 (atom interferometers) and partly 3 (atomic clocks) and 4 (biomedical sensing); does NOT match rows 2 (QKD: awareness and formal both earlier, Phase 1 Conj / Phase 2 U, then Phase 3 H) and 5 (QC applications: awareness and formal both Phase 1, M). Corrections proposed per row. | files: drafts/twodim_check.md | next: U07b drafts/placement_cases.md - 2026-10-25T21:16:51Z | unit=U07b-cases | DONE. drafts/placement_cases.md written: (a) atom interferometry not placed in Phases 1–2 (P1: matter waves only from grade 11, Lehka p.227/p.226; Stadermann Q6), Phase 3 awareness Conj; replaced by the single-photon MZI (Ph1 M, Ph2 U/AM-2 H, Ph3 H) and MZI phase sensing (Ph2 U/AM-2 M, Ph3 M). (b) concept Ph1 Conj → Ph3 H; protocol Ph2 U (AM-2 M) → Ph3 H; complex amplitudes Ph3 only, U (P1–P2 conflict), no t+t record. (c) superposition and entanglement both Phase 1 verbal M (adjacent only), Ph2 U, Ph3 H. (d) matrices Phase 3 optional only, M (adjacent R0056 mixed; P1 only through R0048 [headless] → U without it). ⟦Semerikov et al. (2026) divergences⟧ from ⟦the main model session's notes⟧ (not checked against Table 3). | files: drafts/placement_cases.md | next: U07c drafts/placement_section.md - 2026-10-25T21:18:07Z | unit=U07c-section | DONE. drafts/placement_section.md written (draft text plus a list of manuscript sentences that the matrix contradicts: l.948/967 "most … conjectures" false (33 %); l.982 imprecise; l.823 principle (i) false for D2; l.855 Bouchée diagonal; §3.4 lacks "undecided", Lehka pages 223–233, probability also in profile grade 11). Every number in it was checked against build_report.txt and a recount of grade-only records (16 eligible; 9 inferred; 3 without a named country, which act only through overlap groups; R0056 has a stated eligibility age). | files: drafts/placement_section.md | next: U08 final checks (no source/⟦project log and plan⟧ edits; ⟦the two records excluded from use as evidence⟧ unused), advisor review, handback - 2026-10-25T21:18:41Z | unit=U08-checks | DONE. Section draft trimmed to 482 words. Checks: source/, ⟦project log and plan⟧ not modified (read only); ⟦the two records excluded from use as evidence⟧ absent from all outputs; build asserts both are excluded; rules hash still fd79e952…; placement_units.csv hash unchanged since U03. All deliverables are on disk. | files: drafts/placement_section.md | next: advisor review, then handback - 2026-10-25T21:20:58Z | unit=U09-advisor-fixes | DONE (advisor review before handback). (1) Grid caption now says "age band (reported, or inferred from the grade and the named country)" and "prerequisites, if any", and gives the evidence base as 36 eligible / 25 taught-and-tested / 22 samples. (2) K02-5 Dirac cells now carry R0251 as context: cluster-randomised field studies, ages 15–17; the reduced-Dirac group scored lower on functional fidelity (r = 0.30) and reported higher intrinsic load (r = 0.50). It is not contrary evidence because Dirac notation is only [taught] in R0251 (O7). cell_spec.csv regenerated, sha256 6d95f3ca…. Grades unchanged (H33 M31 L0 Conj40 U19). Grid recompiled: 0 errors, 0 overfull. Rules-literal O4 kept as primary; S-A and S-B are reported per cell in the handback. | files: tools/placement/02_cells.py, tools/placement/cell_spec.csv, tools/placement/build.py, evidence/placement_matrix.csv, drafts/placement_grid*.tex | next: HANDBACK to ⟦the main model session⟧ (T6 complete)

Supplementary S5: Grid as printed in the manuscript (Amendments A1–A4)

Placement of the content of Table 3 across the three phases. Each entry gives a short treatment and, in brackets, the evidence grade of that placement under the rules of Section 3.4: H, learners in the phase's age band (reported, or inferred from the grade and the named country) were taught and tested on the element, its prerequisites, if any, appear in standard curricula by that phase, and no consensus source opposes it; M, evidence only from an adjacent age band, or an opposing consensus; L, contrary evidence; Conj, no evidence from the phase or an adjacent band, so the placement is a conjecture to be tested. Ages are indicative. Of 36 eligible studies with outcomes, 25 taught and tested at least one element; combining reports of the same sample leaves 22 samples. The full matrix (125 placed cells), with sources and page numbers for every cell, is in Supplementary S5. [a]Conditional domains.

Domain	Phase 1 (indicative ages 11–13)	Phase 2 (14–16)	Phase 3 (16–18)
D1 Principles and phenomena	superposition (verbal) [M]; single-photon MZI, qualitative [M]; random single outcomes [H]; uncertainty (verbal) [M]; wave and particle behaviour [M]; correlated pairs (verbal) [M]; no path, no prediction [M]	superposition, real two-path states [H]; single-photon MZI, probabilities [H]; outcome frequencies [H]; single-slit trade-off [H]; photons build interference [H]; pair-outcome probabilities [H]; no-trajectory argument [H]	superposition, real-valued states [H]; MZI, real amplitudes ($0/\pi$) [H]; probabilistic prediction [H]; Heisenberg estimate [H]; de Broglie wavelength [H]; entangled states, real [H]; state vs property [H]
D2 Mathematical representations	–	probabilities as frequencies [H]; real two-state vectors [H]; particle in a box (qualitative) [M]; Dirac notation, real coefficients [M]	probabilities of sequences [H]; real vectors in two bases [H]; particle in a box (qualitative) [M]; complex amplitudes (optional) [Conj]; Dirac notation, real coefficients [H]; 2×2 real matrices (optional) [M]
D3 Atoms, light, and matter	line spectra seen [M]; photons (verbal) [H]; emission, absorption (verbal) [Conj]; semiconductors named [Conj]	level diagrams, $E = hf$ [H]; $E = hf$, photoelectric effect [H]; transitions between levels [M]; energy-gap picture [M]	hydrogen levels, spectra [H]; photoelectric equation [H]; stimulated emission, laser [H]; bands from levels [H]
D4 Information and qubits	bit vs qubit (descriptive) [M]; qubit read-out 0/1 [M]; entanglement as resource (named) [M]; qubit platforms named [M]	qubit as real vector [H]; read-out probabilities [H]; shared key from pairs [H]; what encodes 0 and 1 [H]	states of n qubits [H]; read-out in two bases [H]; teleportation (conceptual) [M]; isolation and cooling [H]
D5 Computing and simulation	what QCs can and cannot do [M]; visual circuit blocks [M]; fragility, decoherence named [M]	which problems gain [H]; X, H, CNOT in a simulator [H]; errors as flipped outcomes [H]	speed-up claims assessed [M]; small circuits (conceptual) [H]; error-correction idea [M]
D6 Communication[a]	eavesdropper leaves traces [Conj]; PQC and QKD named [Conj]; quantum randomness [H]	wrong setting randomises [M]; BB84 walkthrough [M]; how PQC and QKD differ [Conj]; beam-splitter coin [H]	eavesdropper detection [H]; BB84 walkthrough [H]; PQC and QKD compared [Conj]; how a QRNG works [M]
D7 Sensing and metrology[a]	clocks for navigation [Conj]; sensitivity makes sensors [Conj]; sensors and uses named [Conj]	clock from an atomic transition [Conj]; phase sensing with an MZI [M]; what sensors measure [M]; atom interferometry (awareness) [Conj]	clocks from energy levels [Conj]; interference and spin sensing [M]; quantum vs classical sensors [M]; atom interferometry (awareness) [Conj]
D8 Devices and enabling technologies	quantum devices named [M]; photon detectors, scanners [Conj]; cold, vacuum, sources named [Conj]	LEDs, solar cells, $E = hf$ [H]; photodetector principle [M]; why cooling isolates [Conj]	laser principle, bands [M]; coincidence detection, MRI spin [H]; photon sources, cryogenics [Conj]
D9 Nature, history, philosophy	discovery stories [Conj]; disagreement named [Conj]; photon as a model [Conj]; science changes [M]	changing models [Conj]; Schrödinger's cat [Conj]; limits of models [Conj]; determinism revisited [H]	Bohr–Einstein debate [Conj]; interpretations compared [Conj]; models vs reality [Conj]; paradigm change [M]
Practice layer	LEDs, filters, diffraction [Conj]; observe simulations [Conj]; block-based circuits [Conj]	Planck's constant with LEDs [Conj]; test predictions [M]; program small circuits [Conj]	experiments with uncertainty [Conj]; "what if" simulations [H]; high-level toolkit [Conj]
Societal dimension	why encryption must change [Conj]; sensing benefits named [Conj]; "quantum" in adverts [Conj]	PQC migration [Conj]; benefits and limits [Conj]; checking health claims [Conj]	migration trade-offs [Conj]; who benefits [Conj]; pseudoscience analysed [Conj]

Supplementary S5: Grid under the rules as frozen, with the corrected catalogue (A1, A2, A4)

Placement of the content of Table 3 across the three phases. Each entry gives a short treatment and, in brackets, the evidence grade of that placement under the rules of Section 3.4: H, learners in the phase's age band (reported, or inferred from the grade and the named country) were taught and tested on the element, its prerequisites, if any, appear in standard curricula by that phase, and no consensus source opposes it; M, evidence only from an adjacent age band, or an opposing consensus; L, contrary evidence; Conj, no evidence from the phase or an adjacent band, so the placement is a conjecture to be tested; U, undecided, because the rules conflict or the curricular phase of a prerequisite is not documented. Ages are indicative. Of 36 eligible studies with outcomes, 25 taught and tested at least one element; combining reports of the same sample leaves 22 samples. The full matrix (125 placed cells), with sources and page numbers for every cell, is in Supplementary S5. [a]Conditional domains. [Sensitivity version: grades under the rules as frozen (Amendments A1 and A2) with the corrected prerequisite catalogue (Amendment A4), without the post-results Amendment A3; not the primary result.]

Domain	Phase 1 (indicative ages 11–13)	Phase 2 (14–16)	Phase 3 (16–18)
D1 Principles and phenomena	superposition (verbal) [M]; single-photon MZI, qualitative [M]; random single outcomes [H]; uncertainty (verbal) [M]; wave and particle behaviour [M]; correlated pairs (verbal) [M]; no path, no prediction [M]	superposition, real two-path states [H]; single-photon MZI, probabilities [H]; outcome frequencies [H]; single-slit trade-off [H]; photons build interference [H]; pair-outcome probabilities [H]; no-trajectory argument [H]	superposition, real-valued states [H]; MZI, real amplitudes $(0/\pi)$ [H]; probabilistic prediction [H]; Heisenberg estimate [H]; de Broglie wavelength [H]; entangled states, real [H]; state vs property [H]
D2 Mathematical representations	–	probabilities as frequencies [H]; real two-state vectors [H]; particle in a box (qualitative) [M]; Dirac notation, real coefficients [M]	probabilities of sequences [H]; real vectors in two bases [H]; particle in a box (qualitative) [M]; complex amplitudes (optional) [U]; Dirac notation, real coefficients [H]; 2×2 real matrices (optional) [M]
D3 Atoms, light, and matter	line spectra seen [M]; photons (verbal) [H]; emission, absorption (verbal) [Conj]; semiconductors named [Conj]	level diagrams, $E = hf$ [H]; $E = hf$, photoelectric effect [H]; transitions between levels [M]; energy-gap picture [M]	hydrogen levels, spectra [H]; photoelectric equation [H]; stimulated emission, laser [H]; bands from levels [H]
D4 Information and qubits	bit vs qubit (descriptive) [M]; qubit read-out 0/1 [M]; entanglement as resource (named) [M]; qubit platforms named [M]	qubit as real vector [H]; read-out probabilities [H]; shared key from pairs [H]; what encodes 0 and 1 [H]	states of n qubits [H]; read-out in two bases [H]; teleportation (conceptual) [M]; isolation and cooling [H]
D5 Computing and simulation	what QCs can and cannot do [M]; visual circuit blocks [M]; fragility, decoherence named [M]	which problems gain [H]; X, H, CNOT in a simulator [H]; errors as flipped outcomes [H]	speed-up claims assessed [M]; small circuits (conceptual) [H]; error-correction idea [M]
D6 Communication[a]	eavesdropper leaves traces [Conj]; PQC and QKD named [Conj]; quantum randomness [H]	wrong setting randomises [M]; BB84 walkthrough [M]; how PQC and QKD differ [Conj]; beam-splitter coin [H]	eavesdropper detection [H]; BB84 walkthrough [H]; PQC and QKD compared [Conj]; how a QRNG works [M]
D7 Sensing and metrology[a]	clocks for navigation [Conj]; sensitivity makes sensors [Conj]; sensors and uses named [Conj]	clock from an atomic transition [U]; phase sensing with an MZI [M]; what sensors measure [M]; atom interferometry (awareness) [Conj]	clocks from energy levels [Conj]; interference and spin sensing [M]; quantum vs classical sensors [M]; atom interferometry (awareness) [Conj]
D8 Devices and enabling technologies	quantum devices named [M]; photon detectors, scanners [Conj]; cold, vacuum, sources named [Conj]	LEDs, solar cells, $E = hf$ [H]; photodetector principle [M]; why cooling isolates [Conj]	laser principle, bands [M]; coincidence detection, MRI spin [H]; photon sources, cryogenics [Conj]
D9 Nature, history, philosophy	discovery stories [Conj]; disagreement named [Conj]; photon as a model [Conj]; science changes [M]	changing models [Conj]; Schrödinger's cat [Conj]; limits of models [Conj]; determinism revisited [H]	Bohr–Einstein debate [Conj]; interpretations compared [Conj]; models vs reality [Conj]; paradigm change [M]
Practice layer	LEDs, filters, diffraction [Conj]; observe simulations [Conj]; block-based circuits [Conj]	Planck's constant with LEDs [Conj]; test predictions [M]; program small circuits [Conj]	experiments with uncertainty [Conj]; "what if" simulations [H]; high-level toolkit [Conj]
Societal dimension	why encryption must change [Conj]; sensing benefits named [Conj]; "quantum" in adverts [Conj]	PQC migration [Conj]; benefits and limits [Conj]; checking health claims [Conj]	migration trade-offs [Conj]; who benefits [Conj]; pseudoscience analysed [Conj]

Supplementary S5: Grid as first computed (A1–A2, before the A4 correction)

Placement of the content of Table 3 across the three phases. Each entry gives a short treatment and, in brackets, the evidence grade of that placement under the rules of Section 3.4: H, learners in the phase's age band (reported, or inferred from the grade and the named country) were taught and tested on the element, its prerequisites, if any, appear in standard curricula by that phase, and no consensus source opposes it; M, evidence only from an adjacent age band, or an opposing consensus; L, contrary evidence; Conj, no evidence from the phase or an adjacent band, so the placement is a conjecture to be tested; U, undecided, because the rules conflict or the curricular phase of a prerequisite is not documented. Ages are indicative. Of 36 eligible studies with outcomes, 25 taught and tested at least one element; combining reports of the same sample leaves 22 samples. The full matrix (125 placed cells), with sources and page numbers for every cell, is in Supplementary S5. [a]Conditional domains. [Sensitivity version: the grid as first computed, under Amendments A1 and A2 with the prerequisite catalogue before the Amendment A4 correction; not the primary result.]

Domain	Phase 1 (indicative ages 11–13)	Phase 2 (14–16)	Phase 3 (16–18)
D1 Principles and phenomena	superposition (verbal) [M]; single-photon MZI, qualitative [M]; random single outcomes [H]; uncertainty (verbal) [M]; wave and particle behaviour [M]; correlated pairs (verbal) [M]; no path, no prediction [M]	superposition, real two-path states [U]; single-photon MZI, probabilities [U]; outcome frequencies [U]; single-slit trade-off [H]; photons build interference [H]; pair-outcome probabilities [U]; no-trajectory argument [U]	superposition, real-valued states [H]; MZI, real amplitudes ($0/\pi$) [H]; probabilistic prediction [H]; Heisenberg estimate [H]; de Broglie wavelength [H]; entangled states, real [H]; state vs property [H]
D2 Mathematical representations	–	probabilities as frequencies [U]; real two-state vectors [U]; Dirac notation, real coefficients [U]	probabilities of sequences [H]; real vectors in two bases [H]; particle in a box (qualitative) [M]; complex amplitudes (optional) [U]; Dirac notation, real coefficients [H]; 2×2 real matrices (optional) [M]
D3 Atoms, light, and matter	line spectra seen [M]; photons (verbal) [H]; emission, absorption (verbal) [Conj]; semiconductors named [Conj]	level diagrams, $E = hf$ [H]; $E = hf$, photoelectric effect [H]; transitions between levels [M]; energy-gap picture [M]	hydrogen levels, spectra [H]; photoelectric equation [H]; stimulated emission, laser [H]; bands from levels [H]
D4 Information and qubits	bit vs qubit (descriptive) [M]; qubit read-out 0/1 [M]; entanglement as resource (named) [M]; qubit platforms named [M]	qubit as real vector [U]; read-out probabilities [U]; shared key from pairs [U]; what encodes 0 and 1 [H]	states of n qubits [H]; read-out in two bases [H]; teleportation (conceptual) [M]; isolation and cooling [H]
D5 Computing and simulation	what QCs can and cannot do [M]; visual circuit blocks [M]; fragility, decoherence named [M]	which problems gain [H]; X, H, CNOT in a simulator [U]; errors as flipped outcomes [U]	speed-up claims assessed [M]; small circuits (conceptual) [H]; error-correction idea [M]
D6 Communication[a]	eavesdropper leaves traces [Conj]; PQC and QKD named [Conj]; quantum randomness [H]	wrong setting randomises [U]; BB84 walkthrough [U]; how PQC and QKD differ [Conj]; beam-splitter coin [U]	eavesdropper detection [H]; BB84 walkthrough [H]; PQC and QKD compared [Conj]; how a QRNG works [M]
D7 Sensing and metrology[a]	clocks for navigation [Conj]; sensitivity makes sensors [Conj]; sensors and uses named [Conj]	clock from an atomic transition [U]; phase sensing with an MZI [U]; what sensors measure [M]	clocks from energy levels [Conj]; interference and spin sensing [M]; quantum vs classical sensors [M]; atom interferometry (awareness) [Conj]
D8 Devices and enabling technologies	quantum devices named [M]; photon detectors, scanners [Conj]; cold, vacuum, sources named [Conj]	LEDs, solar cells, $E = hf$ [H]; photodetector principle [M]; why cooling isolates [Conj]	laser principle, bands [M]; coincidence detection, MRI spin [H]; photon sources, cryogenics [Conj]
D9 Nature, history, philosophy	discovery stories [Conj]; disagreement named [Conj]; photon as a model [Conj]; science changes [M]	changing models [Conj]; Schrödinger's cat [Conj]; limits of models [Conj]; determinism revisited [H]	Bohr–Einstein debate [Conj]; interpretations compared [Conj]; models vs reality [Conj]; paradigm change [M]
Practice layer	LEDs, filters, diffraction [Conj]; observe simulations [Conj]; block-based circuits [Conj]	Planck's constant with LEDs [Conj]; test predictions [M]; program small circuits [Conj]	experiments with uncertainty [Conj]; "what if" simulations [H]; high-level toolkit [Conj]
Societal dimension	why encryption must change [Conj]; sensing benefits named [Conj]; "quantum" in adverts [Conj]	PQC migration [Conj]; benefits and limits [Conj]; checking health claims [Conj]	migration trade-offs [Conj]; who benefits [Conj]; pseudoscience analysed [Conj]